

CSc 360
Operating Systems
File Systems

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Review

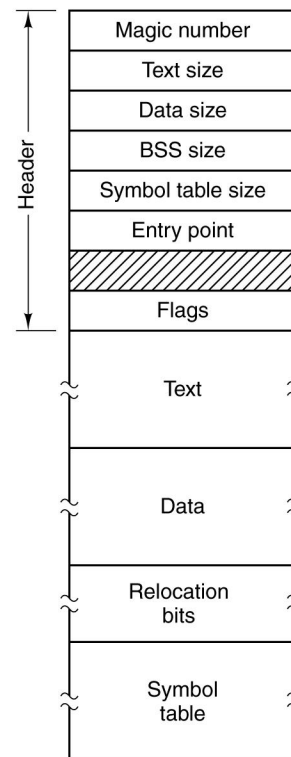
- Operating systems
 - process management
 - how to share CPU: jobs, processes, threads
 - scheduling, communication, synchronization
- What's next?
 - **file systems (this and next week)**
 - memory management
 - I/O systems

File concepts

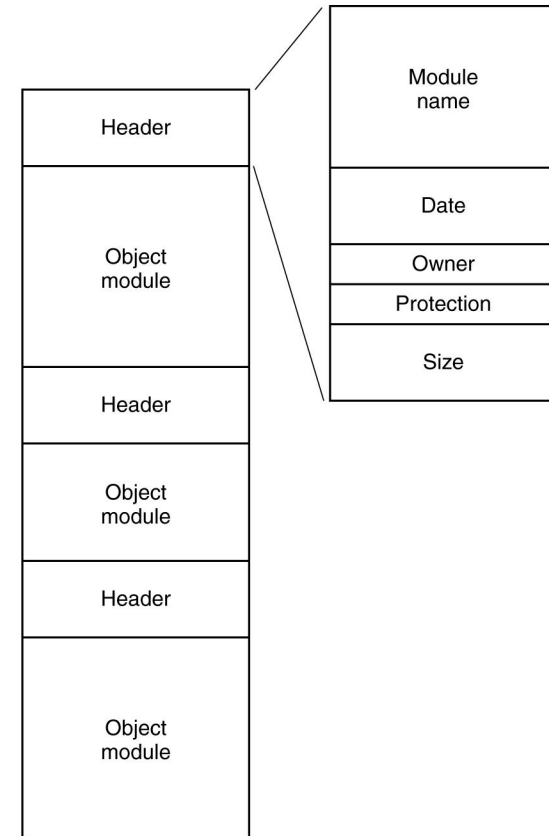
- File: logical storage unit for information
 - normally on disk and tape: nonvolatile
- File type: data (text, binary, ...), program
- File attributes: meta information
 - name, size, time, owner, protection, location, ...
 - kept in directory (a special file)
- File operations
 - create, write, read, seek, append, truncate, etc

File structures

- How information organized within a file
 - unstructured, or
 - structured w/ records
- Example
 - binary executable
 - binary archive
 - modules



(a)



(b)

File access

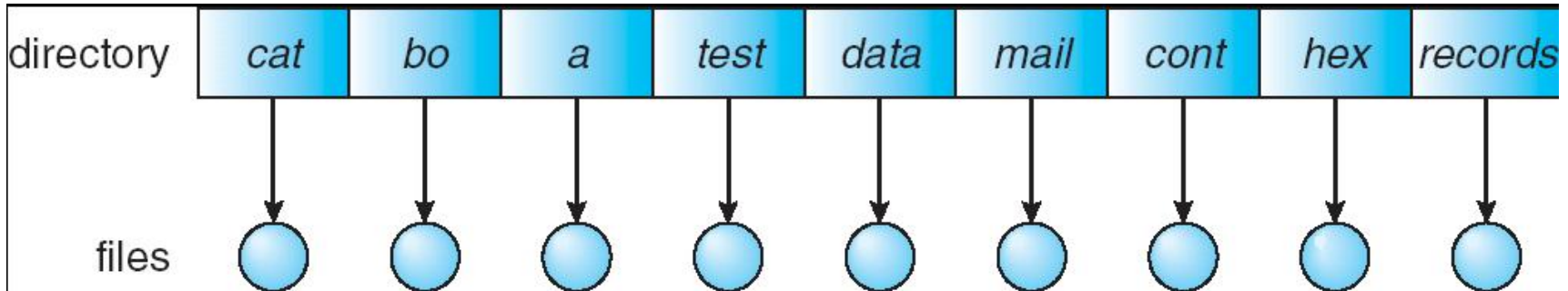
- Sequential access
 - variable length records, i.e., $r_1, r_2, \dots, r_n$
 - to access record r_i , go through r_1 to r_{i-1} first
- Direct access
 - fixed length records of size s
 - i -th record: offset $(i-1)*s$
- Indexed access
 - access index first

Directory concepts

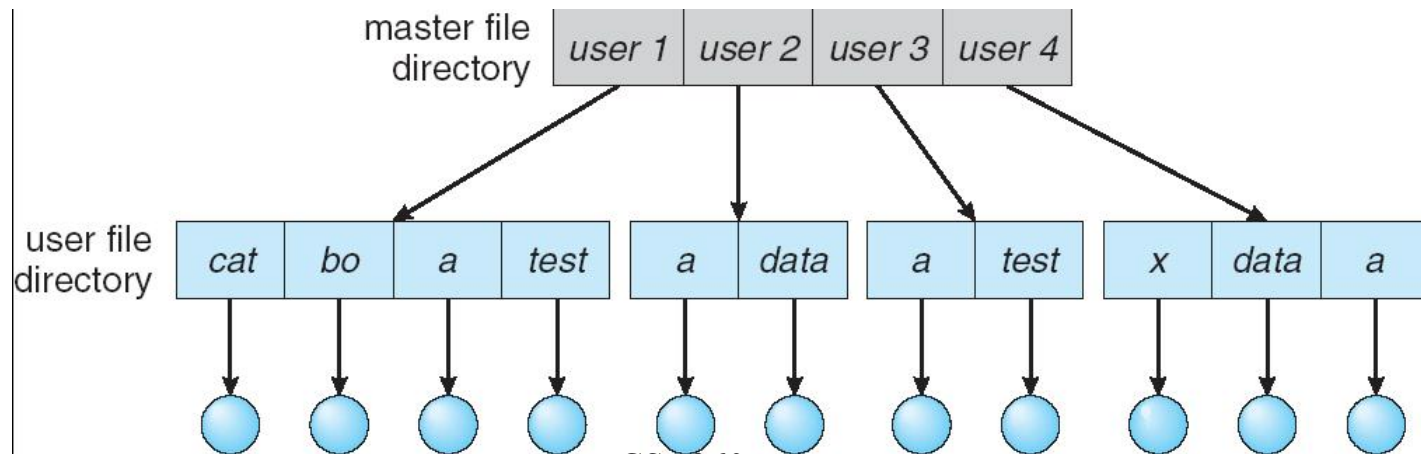
- A special file
 - a collection of pointers to meta information of files
 - normally structured
 - and allow direct access
- Directory operations
 - “file” operations on directory records
- File systems on disk
 - partitions, file systems (FDT/FAT), files

Directory structures

- Single-level directory

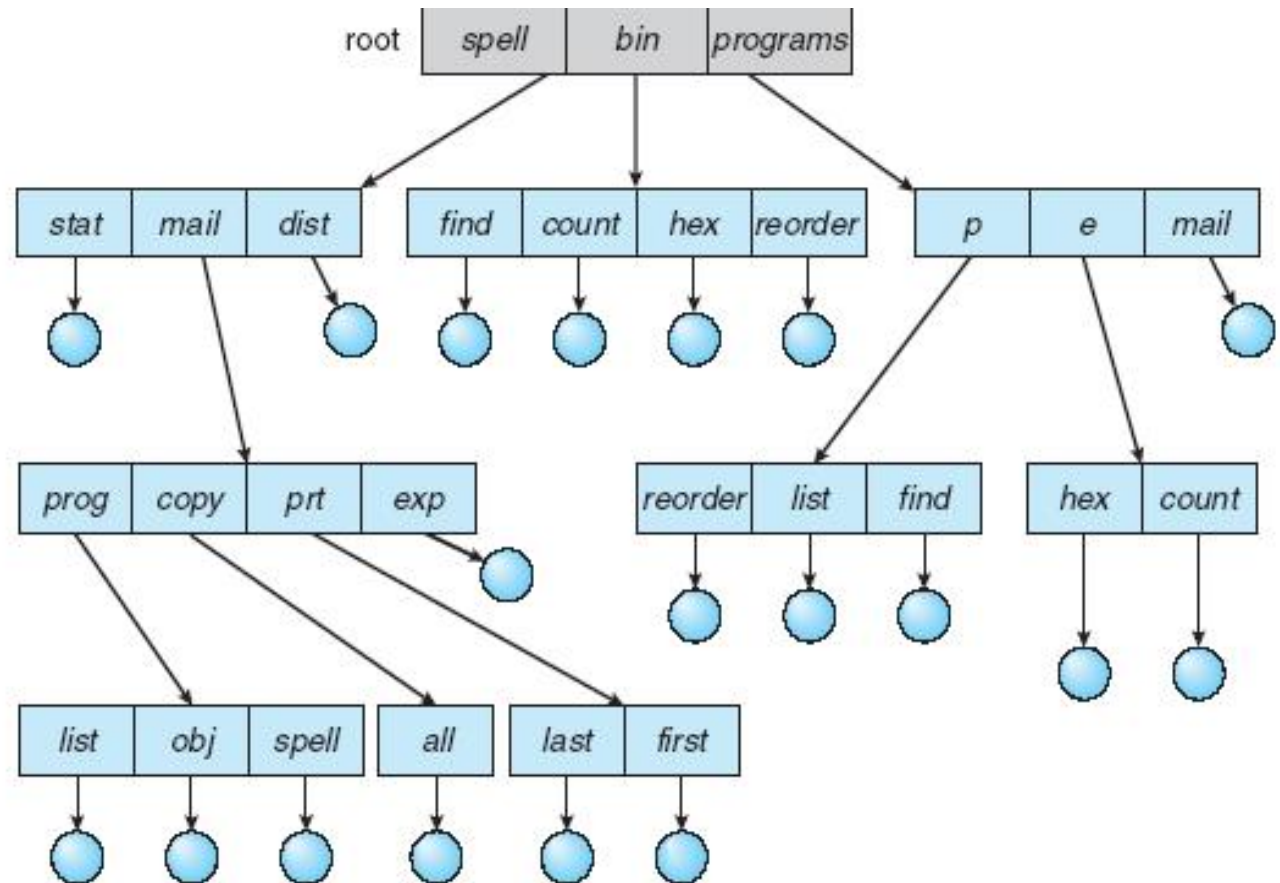


- Two-level directory



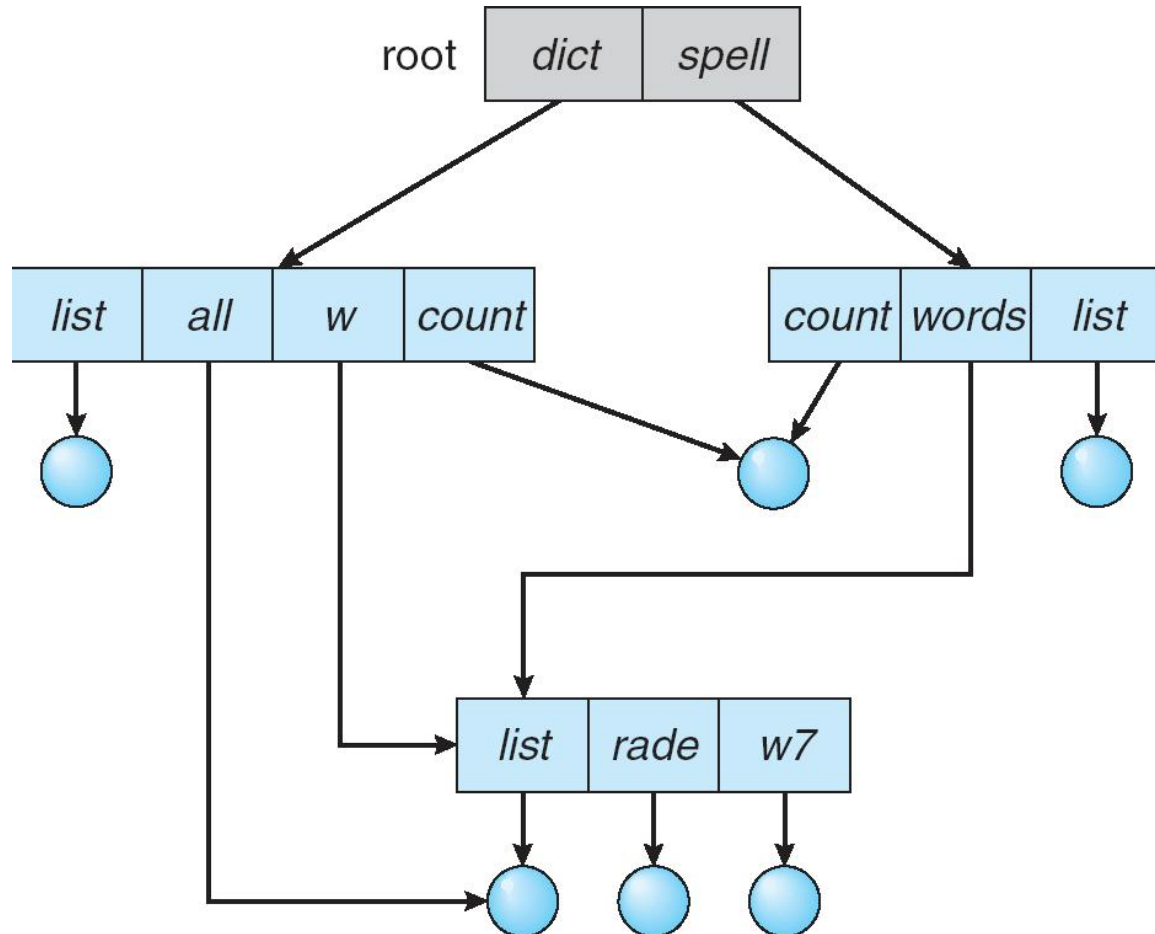
Directory structures: tree

- Path
 - absolute
 - relative



Directory structures: graph

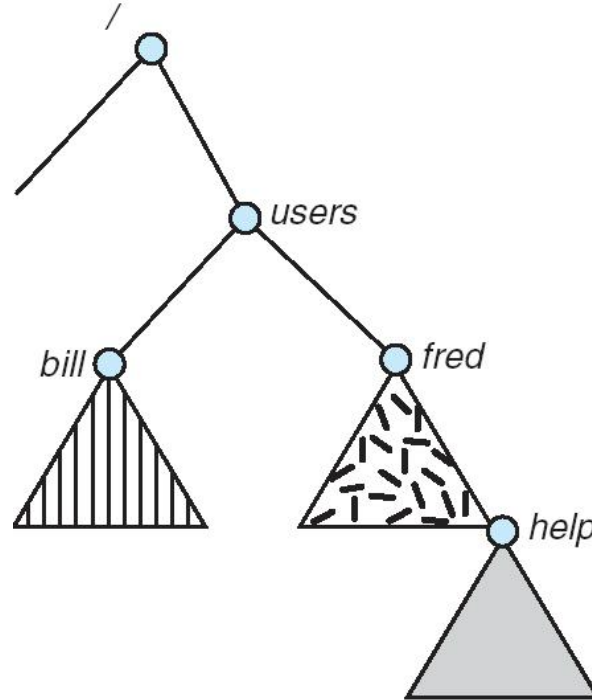
- Links
 - hard links
 - symbolic links
- Link operations
 - resolve links
 - delete links/files
- Acyclic-graph
 - loop-free



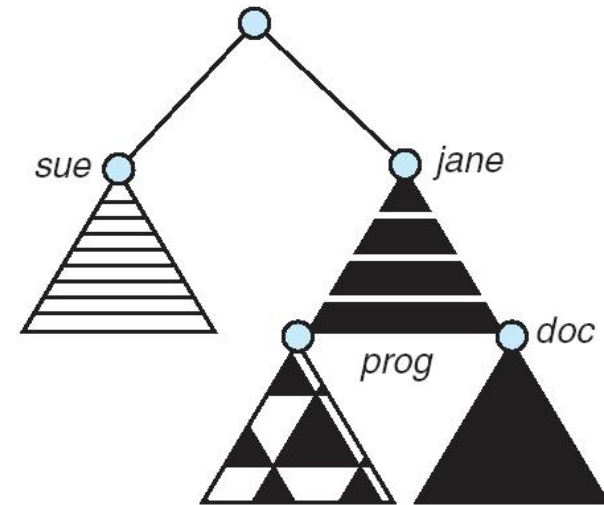
File system operations

- Mount file systems

- mount point
 - directory
 - directory with files?
- multi-mount



(a)
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(b)
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Share and protection

- File attributes
 - owner: user ID, group ID
- Type of access
 - read, write, execute, etc
- Share over the network
 - host ID
 - NFS: network file systems
 - remote procedure calls

Failure mode

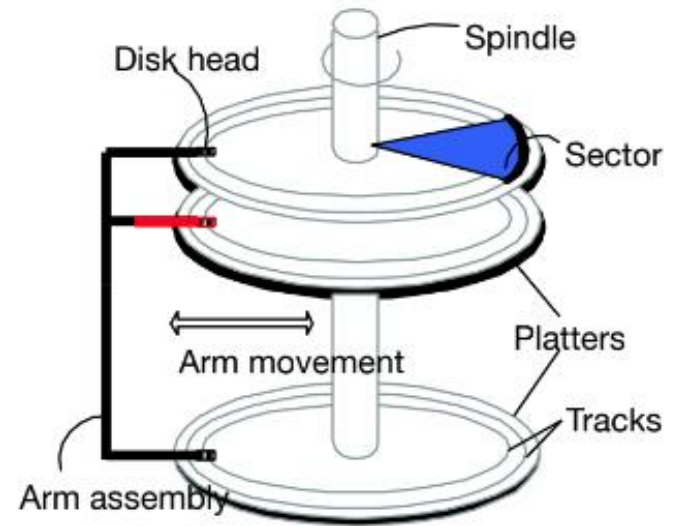
- Local file systems
 - hardware failure
 - corrupted information
- remote file systems
 - network failure
- Fault-tolerant systems
 - keep state at both sides, or
 - stateless but less secure

Consistency control

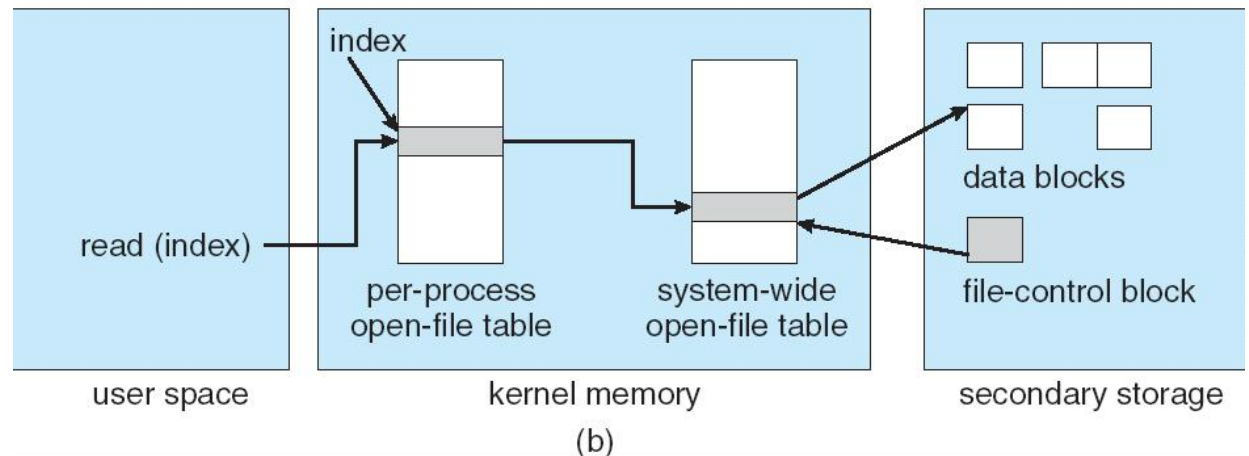
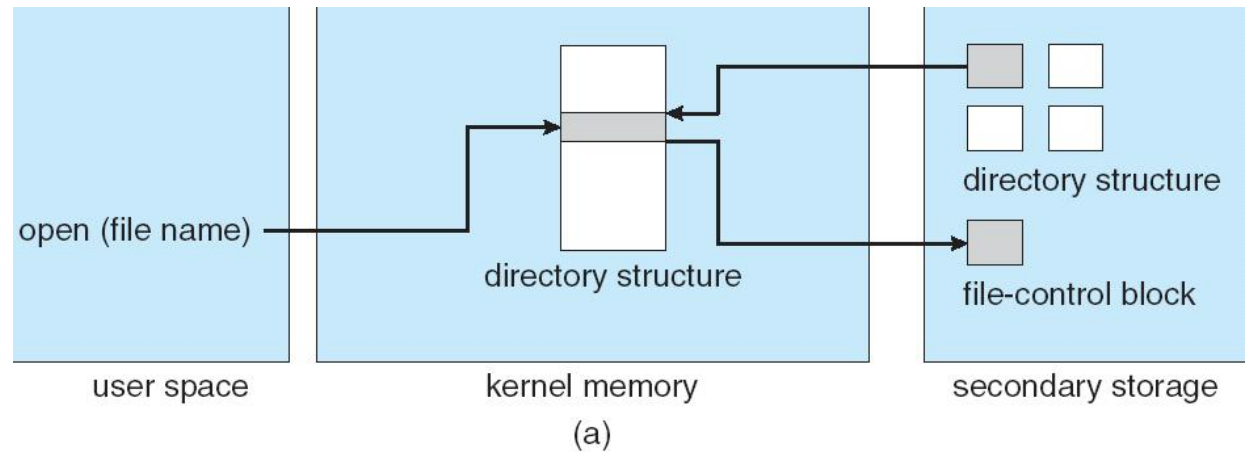
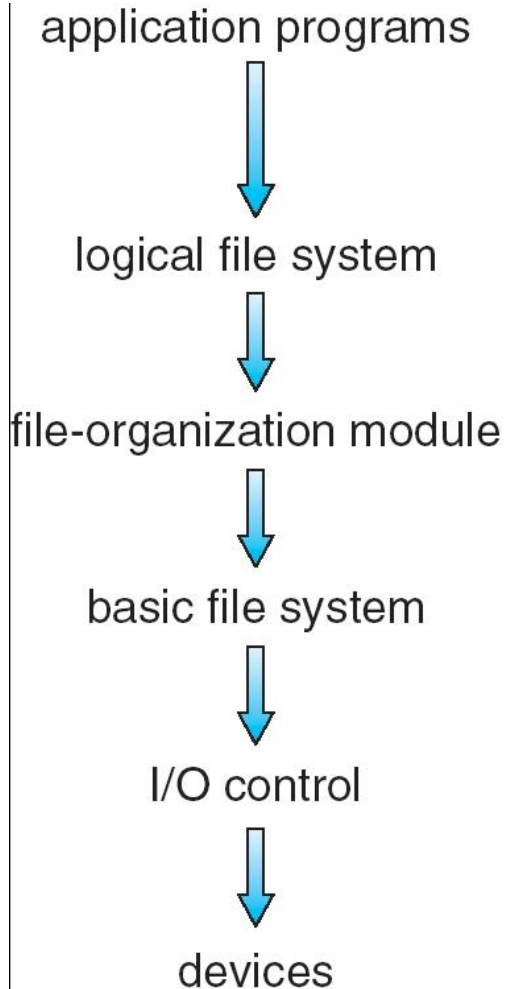
- Multiple users access a file simultaneously
 - lock!
 - entire file, or
 - certain records
 - complexity and (network) overhead
- Consistency semantics
 - Unix semantics: write immediately visible
 - session (open()-close()) semantics: e.g., AFS

File system organization

- Sector: e.g., 512 bytes
- Block
 - data blocks
 - control blocks
 - boot, partition, (root) directory, file control blocks
- Access blocks
 - logical block address
 - physical block address: disk geometry

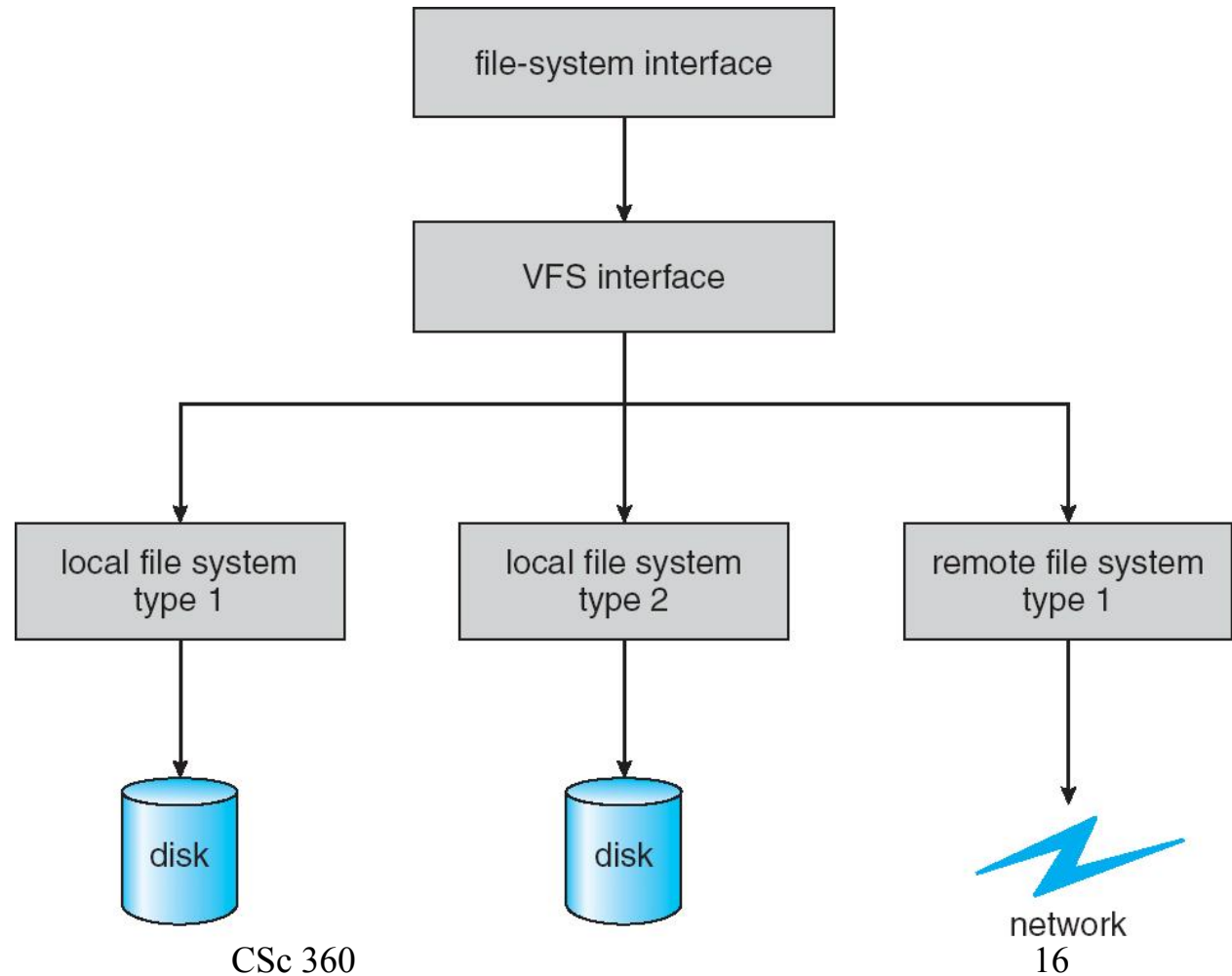


File system structures



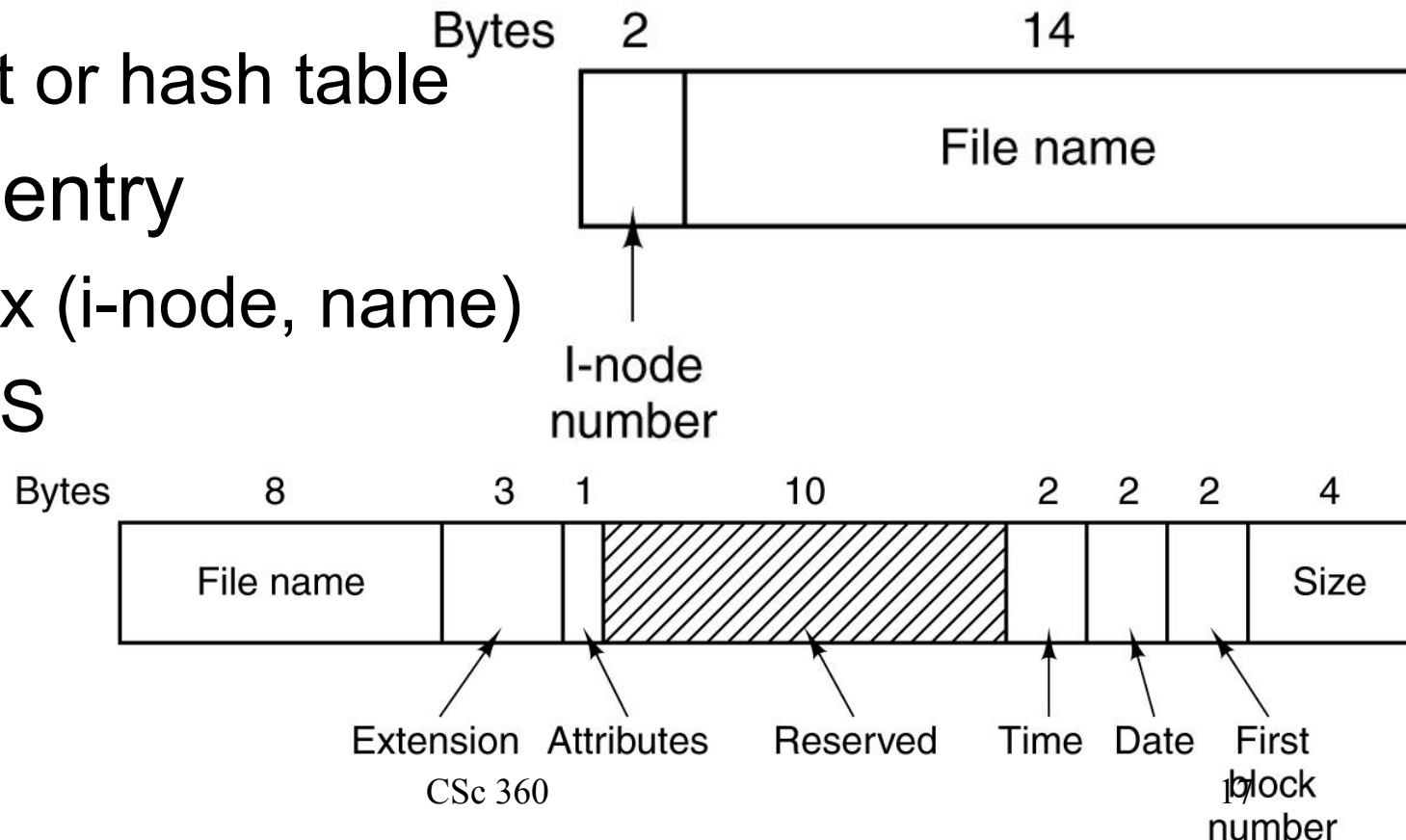
Virtual file systems

- OO design
 - same API to different file systems
 - e.g., open(), read(), write(), etc



Implementing directory

- Directory table
 - linear list or hash table
- Directory entry
 - e.g., Unix (i-node, name)
 - e.g., DOS



Traversing directory hierarchy

- Unix: /usr/ast/mbox

Root directory

1	.
1	..
4	bin
7	dev
14	lib
9	etc
6	usr
8	tmp

Looking up
usr yields
i-node 6

I-node 6
is for /usr

Mode size times
132

I-node 6
says that
/usr is in
block 132

Block 132
is /usr
directory

6	.
1	..
19	dick
30	erik
51	jim
26	ast
45	bal

/usr/ast
is i-node
26
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I-node 26
is for
/usr/ast

Mode size times
406

I-node 26
says that
/usr/ast is in
block 406

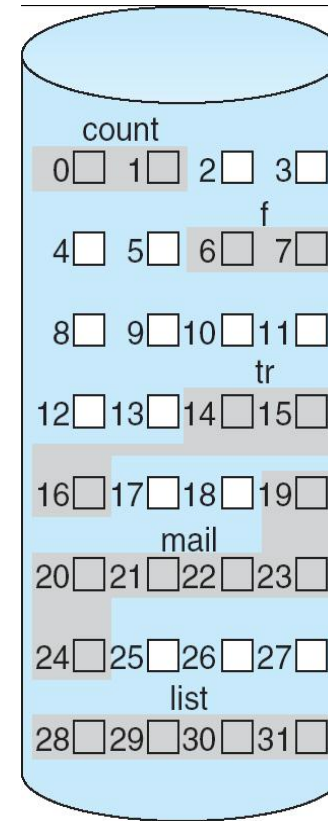
Block 406
is /usr/ast
directory

26	.
6	..
64	grants
92	books
60	mbox
81	minix
17	src

/usr/ast/mbox
is i-node
60

Allocating blocks

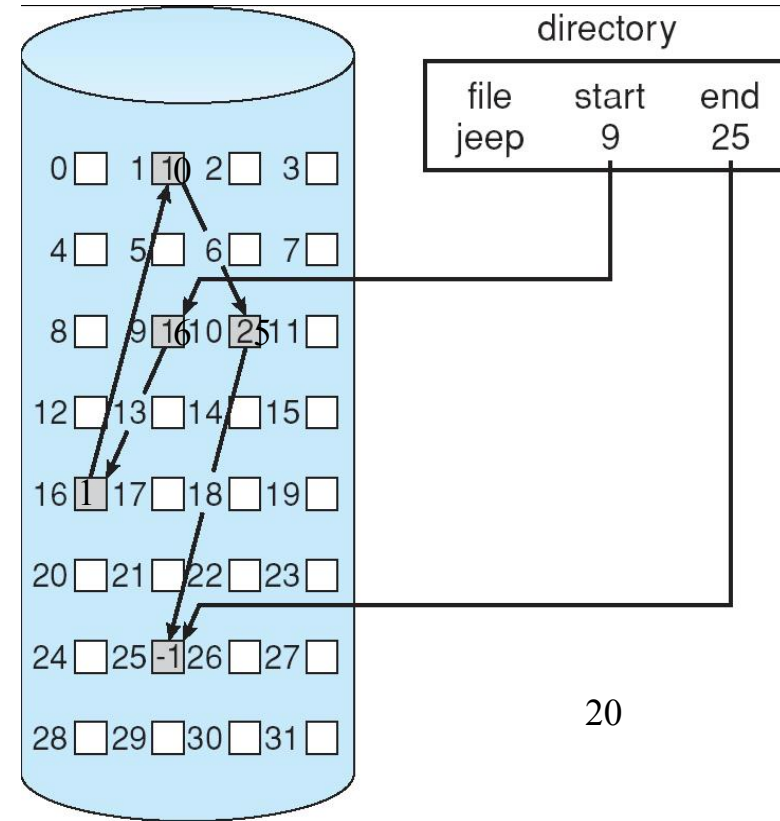
- Contiguous allocation
 - address of the first block, length
 - random access
 - difficult to expand
 - fragments
- Modified contiguous allocation
 - e.g., extent-based allocation



directory		
file	start	length
count	0	2
tr	14	3
mail	19	6
list	28	4
f	6	2

Linked allocation

- Linked list
 - address of the first block
 - each block has a next pointer
 - next == -1, end of block list
 - easy to expand
 - no direct random access
- Modified linked list
 - e.g., cluster-based allocation



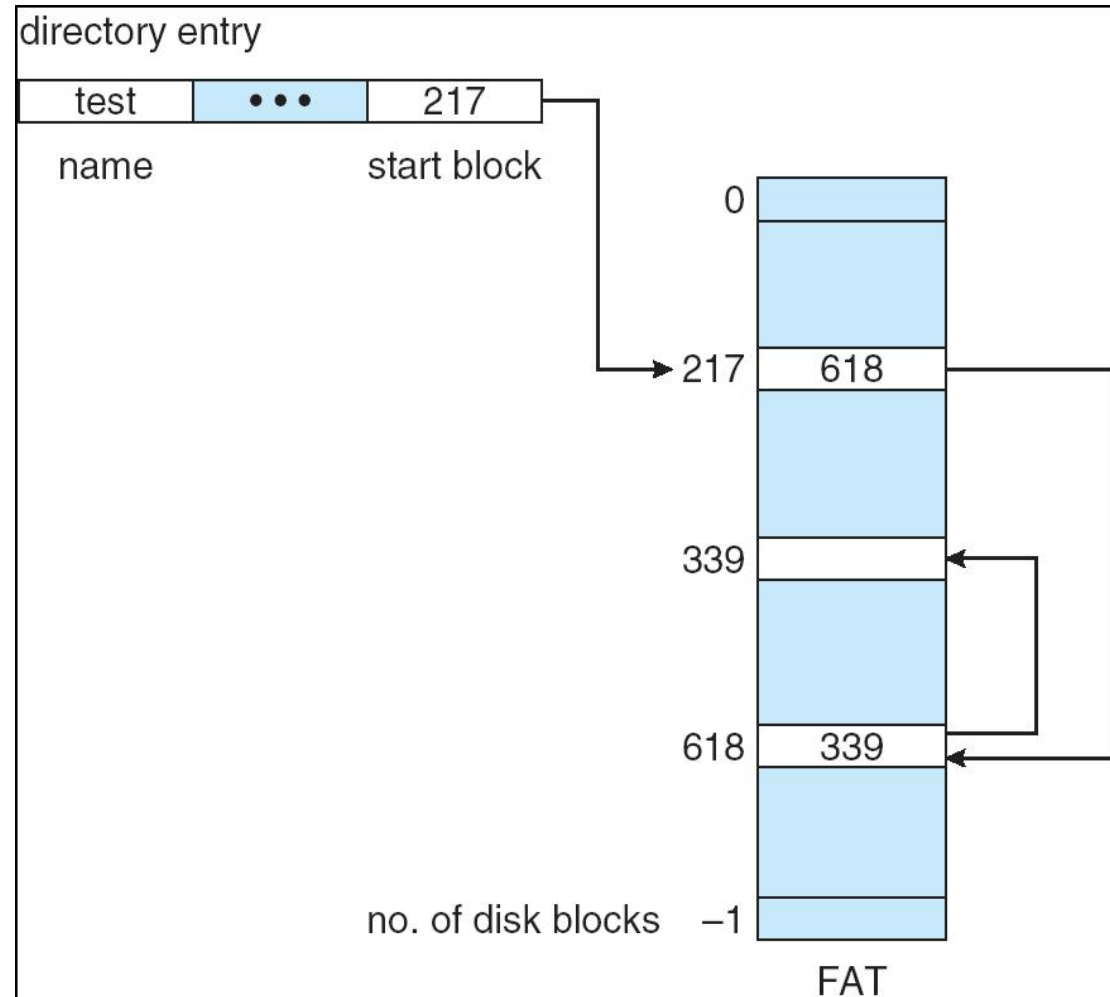
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* 9 -> 16 -> 1 -> 10 -> 25 -> -1 (EOF)

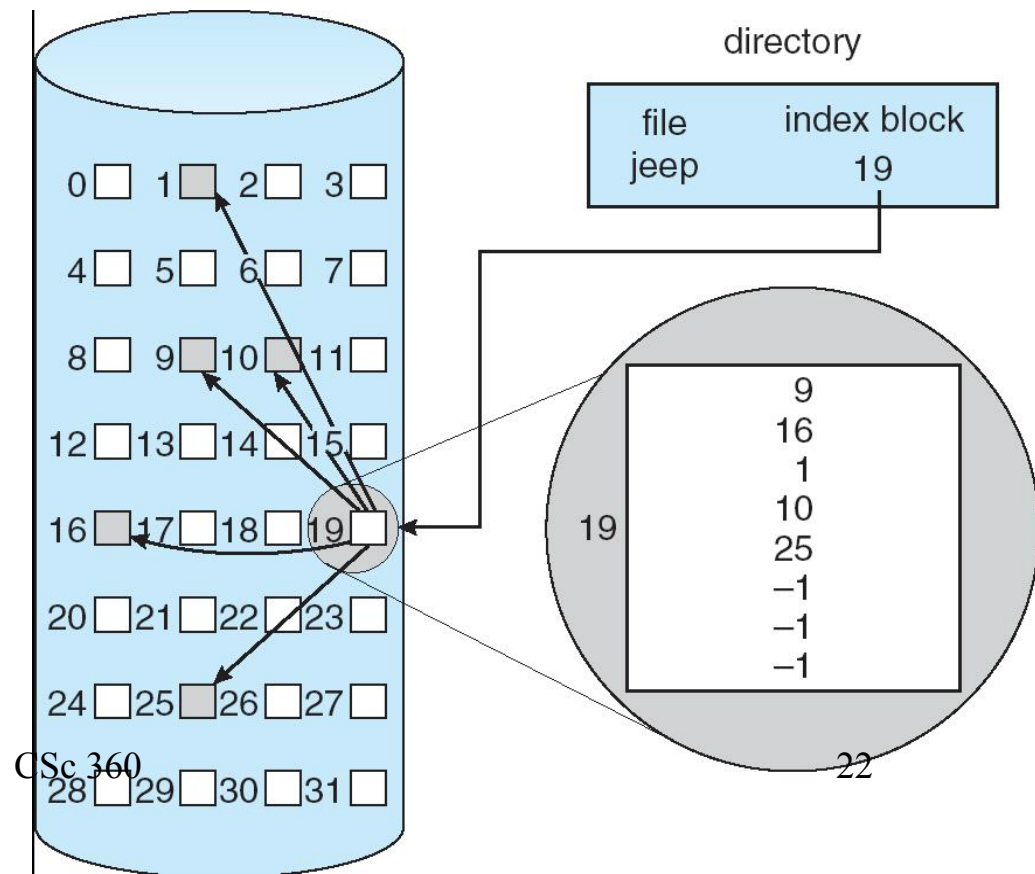
File allocation table

- FDT
 - first block address
- FAT
 - linked list of addresses
 - can be cached
 - random access



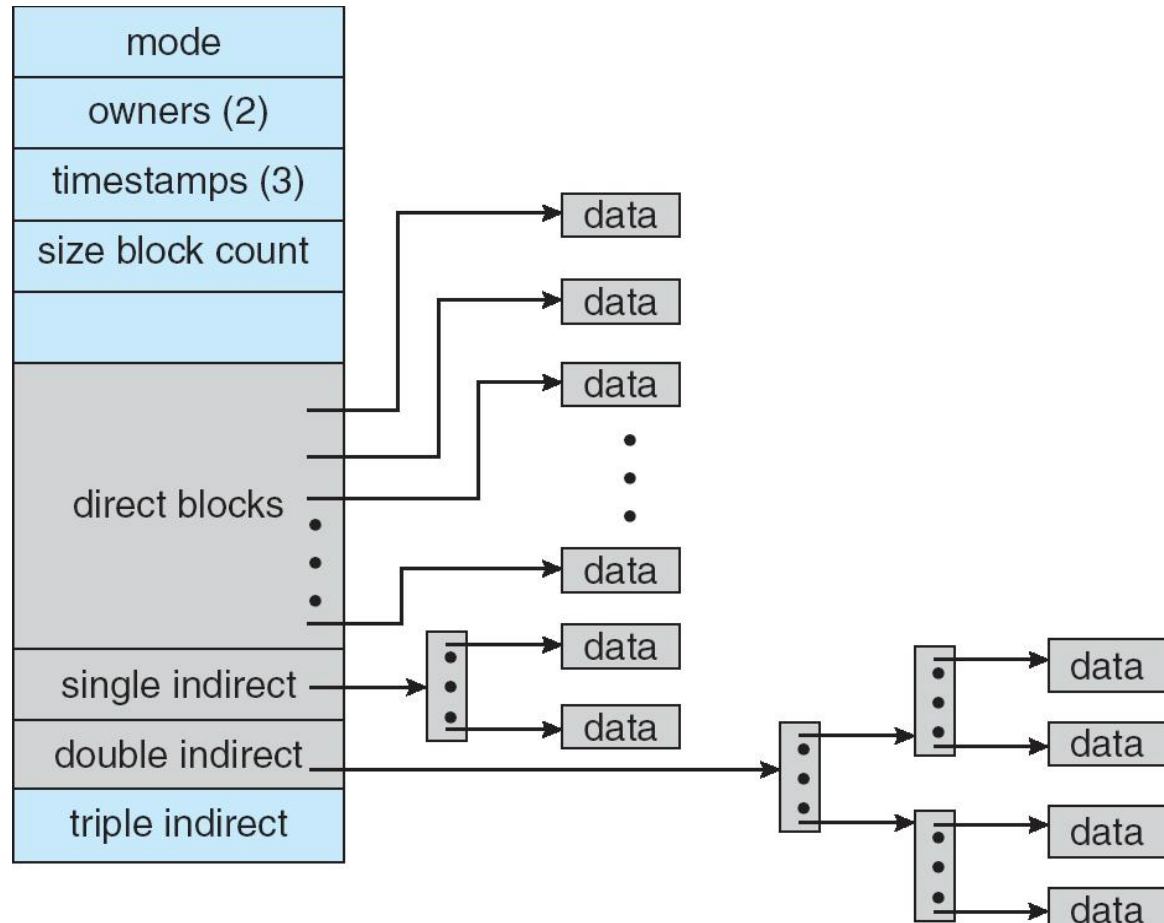
Indexed allocation

- Index block
 - a list of pointers
 - random access
 - easy to expand
 - limited by the size of index block



Indirect index

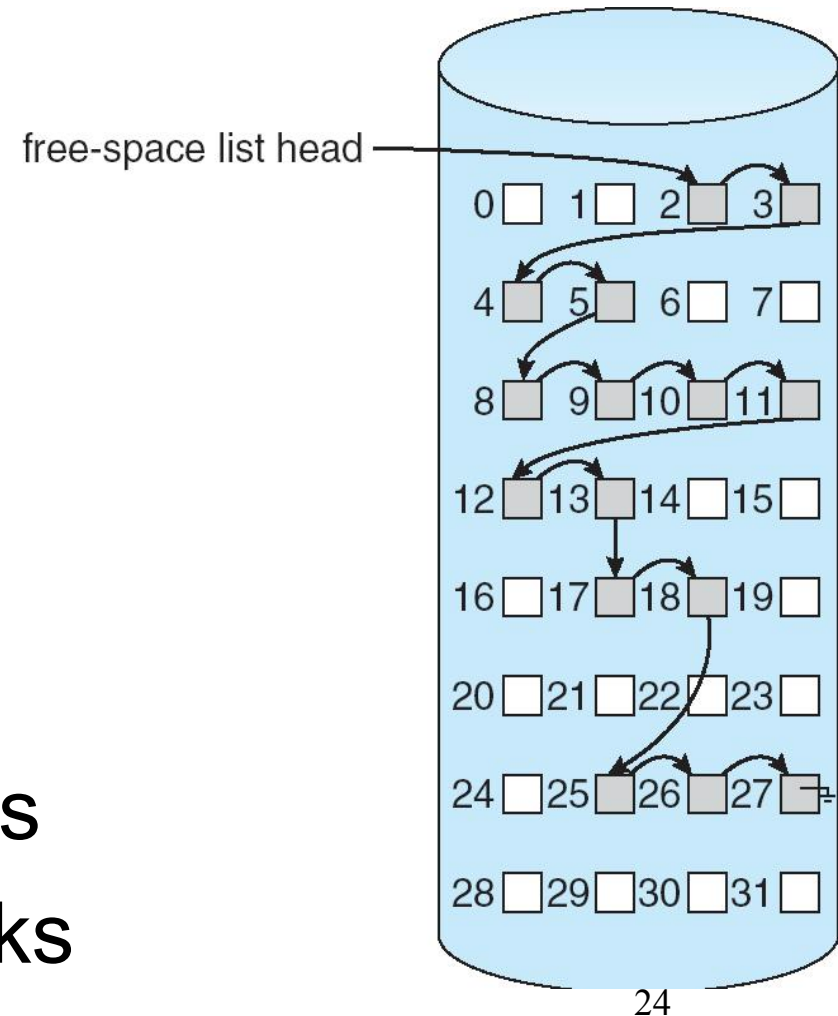
- E.g., Unix i-node
 - 12 direct index
 - 1 single indirect
 - 1 double indirect
 - 1 triple indirect
 - 4KB block



* every cs problem can be solved by another layer of indirection ;-)

Free space management

- Keep track of free blocks
- Bitmap
 - 1 bit indicates whether a block is free or not
- Linked-list
 - “a big free file”
- Grouping: list of free blocks
- Counting: contiguous blocks



* How does the FAT system do this?

Efficiency and performance

- Efficiency: tradeoff
 - e.g., FDT/FAT/block size, # of blocks/entries
 - static vs dynamic data structure
- Performance: further tradeoff
 - buffer vs page cache
 - unified buffer cache
 - free-behind, read-ahead: sequential access
 - RAM disk

Consistency check

- Inconsistency
 - between files and their meta-information
 - due to hardware/software failure
- Consistency check
 - e.g., fsck: file systems consistency check
 - superblock consistency
 - wrong counters
 - inode consistency
 - unreferenced inodes, missing free blocks, bad *free* blocks

Backup and restore

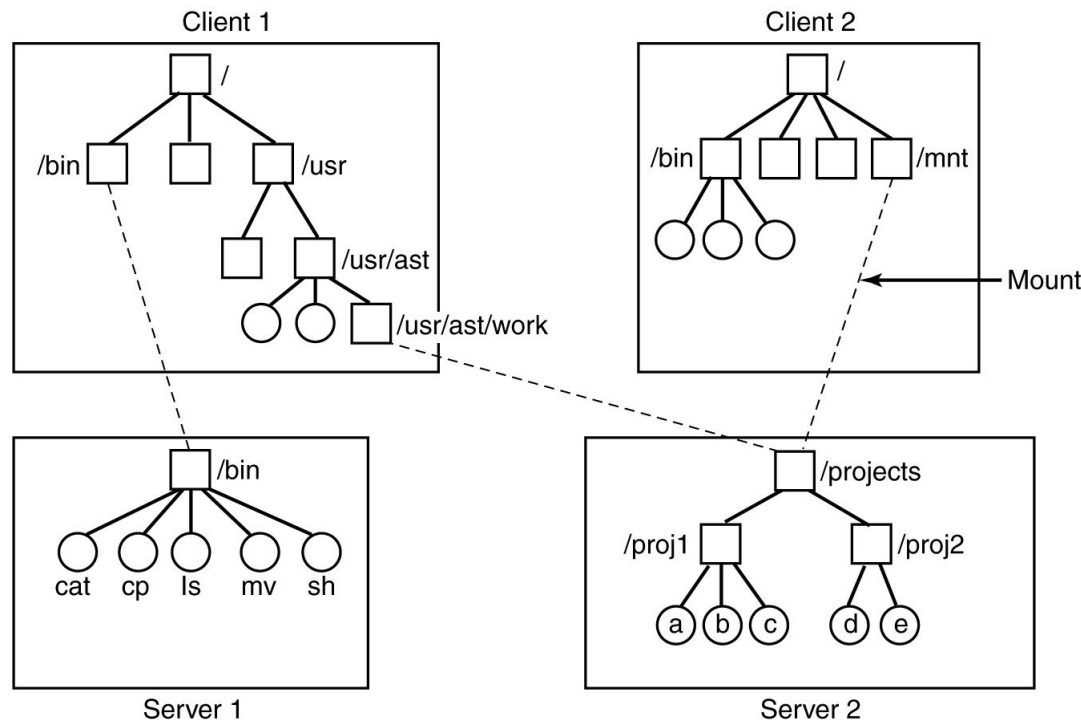
- Backup to another medium
 - magnetic, optical, etc
- Full backup
- Incremental backup
 - last backup time
 - last modify time
- Backup cycle
 - full backup, {incremental backup}*
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Log-structured file system

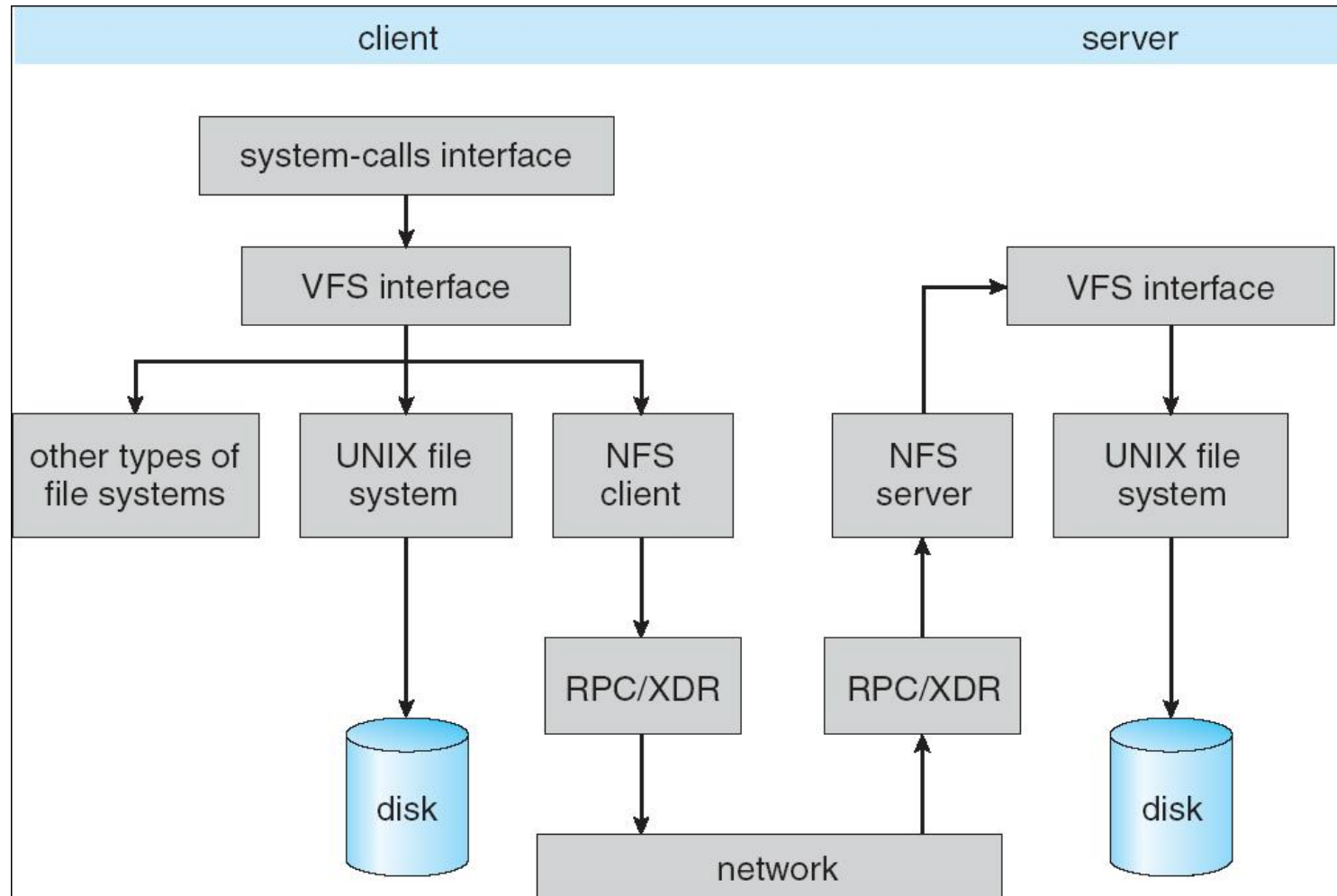
- Transactions: updates to file system
 - considered committed when written to a log
 - file system may not be updated yet
 - asynchronously update file system
 - performance improvement as well
- After system crashes
 - transactions still to be completed
 - file system remains consistency

SunOS NFS

- Access remote file system
- Initially on UDP
- With RPC/XDR
- Transparency
- Mount protocol
 - server: exportfs
 - access control
 - client: mount server:dir dir



NFS architecture



Linux file systems

- MINIX
 - up to 64MB system size, 14B file name
- Journal file system (JFS), XFS, etc
- Extended file system (ext)
 - ext, 1992, up to 2GB system, 255B name
 - ext2, 1993, up to 4TB system (1KB block)
 - ext3, since 1998, ext2 + journaling
 - journal (low risk), ordered, writeback (high)
 - ext4, since 2008, 16TB (file), 1EB (system)

These lectures

- File system design and implementation
 - file system structures
 - directory implementation
 - block allocation
 - free-space management
 - consistency and recovery
- Explore further
 - /bin/mount; /sbin/fsck; /etc/exports

Next lectures

- Memory management
 - Main memory
 - Virtual memory